

Introduction to Abstract Algebra III: Homework 8

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Exercise 47.5. Describe the group of the polynomial $(x^5 - 2) \in (\mathbb{Q}(\zeta))[x]$ over $\mathbb{Q}(\zeta)$, where ζ is a primitive 5th root of unity.

Solution. Let $\alpha = \sqrt[5]{2}$. We start by finding the degree of the splitting field of $f(x) = x^5 - 2$ over $\mathbb{Q}$. Notice that the roots of $f(x)$ are given by $\alpha, \zeta\alpha, \zeta^2\alpha, \zeta^3\alpha, \zeta^4\alpha$. Since ζ is a primitive 5th root of unity, the splitting field of $f(x)$ over $\mathbb{Q}$ is $K = \mathbb{Q}(\alpha, \zeta)$. Now, we compute the degree of $[K : \mathbb{Q}]$. Using the tower law, we have:

$$[K : \mathbb{Q}] = [K : \mathbb{Q}(\zeta)][\mathbb{Q}(\zeta) : \mathbb{Q}].$$

We start with $[K : \mathbb{Q}(\zeta)]$. Since $\alpha \notin \mathbb{Q}(\zeta)$ (as that would mean that α is a root of a polynomial of degree less than 5, which contradicts the irreducibility of $f(x)$), we have $[K : \mathbb{Q}(\zeta)] = 5$. Next, we compute $[\mathbb{Q}(\zeta) : \mathbb{Q}]$. The field $\mathbb{Q}(\zeta)$ is generated by a primitive 5th root of unity, and the minimal polynomial of ζ over $\mathbb{Q}$ is the 5th cyclotomic polynomial, which has degree $\varphi(5) = 4$. Thus, $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 4$. Therefore, we have:

$$[K : \mathbb{Q}] = 5 \cdot 4 = 20.$$

Thus, the Galois group of the splitting field of $f(x)$ over $\mathbb{Q}$ is a group of order 20. Therefore, we have $|\text{Gal}(K/\mathbb{Q}(\zeta))| = 20/4 = 5$. Since 5 is prime, there is only one group of order 5, which is $\mathbb{Z}_5$. Thus,

$$\text{Gal}(K/\mathbb{Q}(\zeta)) \cong \mathbb{Z}_5. \quad \square$$

Exercise 47.8. Find the splitting field K in $\mathbb{C}$ of the polynomial $(x^4 - 4x^2 - 1) \in \mathbb{Q}[x]$. Compute the group of the polynomial over $\mathbb{Q}$ and exhibit the correspondence between the subgroups of $G(K/\mathbb{Q})$ and the intermediate fields. In other words, do the complete job.

Solution. Let $y = x^2$. Thus, we have $f(x) = x^4 - 4x^2 - 1 = y^2 - 4y - 1$. Solving for y , we get:

$$y = 2 \pm \sqrt{5}.$$

Thus, the roots of $f(x)$ are given by $\pm\sqrt{2+\sqrt{5}}$ and $\pm\sqrt{2-\sqrt{5}}$. Let $\alpha = \sqrt{2+\sqrt{5}}$ and $\beta = \sqrt{2-\sqrt{5}}$. We start by finding the degree of the splitting field of $f(x)$ over $\mathbb{Q}$. Computing $\alpha\beta = i$. Thus, we have $\beta = i/\alpha$, meaning $K = \mathbb{Q}(\alpha, i)$. Now, we compute the degree of $[K : \mathbb{Q}]$. Using the tower law, we have:

$$[K : \mathbb{Q}] = [K : \mathbb{Q}(\alpha)][\mathbb{Q}(\alpha) : \mathbb{Q}].$$

We start with $[K : \mathbb{Q}(\alpha)]$. Since $\alpha \in \mathbb{R}$, every element of $\mathbb{Q}(\alpha)$ is also in $\mathbb{R}$. Thus, $i \notin \mathbb{Q}(\alpha)$. Thus, we have $[K : \mathbb{Q}(\alpha)] = 2$. Next, we compute $[\mathbb{Q}(\alpha) : \mathbb{Q}]$. The field $\mathbb{Q}(\alpha)$ is generated by α , and the minimal polynomial of α over $\mathbb{Q}$ is $f(x)$, which has degree 4. Thus, $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. Therefore, we have:

$$[K : \mathbb{Q}] = 2 \cdot 4 = 8.$$

There are 5 groups of order 8. The abelian groups of order 8 are $\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ and the non-abelian groups are D_4 , and Q_8 . We start by constructing all 8 automorphisms.

$$\begin{aligned} \sigma_1 : \alpha &\mapsto \alpha, & i &\mapsto i \\ \sigma_2 : \alpha &\mapsto -\alpha, & i &\mapsto i \\ \sigma_3 : \alpha &\mapsto i\alpha, & i &\mapsto i \\ \sigma_4 : \alpha &\mapsto -i\alpha, & i &\mapsto i \\ \sigma_5 : \alpha &\mapsto \alpha, & i &\mapsto -i \\ \sigma_6 : \alpha &\mapsto -\alpha, & i &\mapsto -i \\ \sigma_7 : \alpha &\mapsto i\alpha, & i &\mapsto -i \\ \sigma_8 : \alpha &\mapsto -i\alpha, & i &\mapsto -i. \end{aligned}$$

We know that $G(K/\mathbb{Q})$ isn't abelian, since $\sigma_2\sigma_3(\alpha) = -i\alpha$ but $\sigma_3\sigma_2(\alpha) = i\alpha$. Thus, we have $G(K/\mathbb{Q}) \cong D_4$.

Now, we compute all possible subgroups of D_4 . Let $\rho = \sigma_3$ and $\mu = \sigma_5$. From Lagrange's Theorem, we know that the order of the subgroup must divide 8, meaning it's either 1, 2, 4, or 8. There is only one subgroup of order 1, which is the trivial subgroup, $\{e\}$ and one subgroup of order 8, which is D_4 itself. All possible subgroups are given by:

$$\begin{aligned} H_1 = \langle \mu \rangle, \quad H_2 = \langle \rho^2 \rangle, \quad H_3 = \langle \mu\rho \rangle, \quad H_4 = \langle \mu\rho^2 \rangle, \quad H_5 = \langle \mu\rho^3 \rangle, \\ H_6 = \langle \rho \rangle, \quad H_7 = \langle \rho^2, \mu \rangle, \quad H_8 = \langle \rho^2, \mu\rho^2 \rangle \end{aligned}$$

Now, we compute the fixed fields of each subgroup. Let $\gamma = a + b\alpha + c\alpha^2 + d\alpha^3 + ei + f\alpha i + g\alpha^2 i + h\alpha^3 i$.

- K_{H_1} : Computing $\mu(\gamma)$, the only unchanged term is $a + b\alpha + c\alpha^2 + d\alpha^3$. Thus, the fixed field $K_{H_1} = \{x \in K \mid \mu(x) = x\}$ is given by $\mathbb{Q}(\alpha)$.
- K_{H_2} : Computing $\rho^2(\gamma)$, the only unchanged term is $a + c\alpha^2 + ei + g\alpha^2 i$. Thus, the fixed field $K_{H_2} = \{x \in K \mid \rho^2(x) = x\}$ is given by $\mathbb{Q}(\alpha^2, i)$.
- K_{H_3} : Computing $\mu\rho(\gamma)$, the only unchanged term is $a + c\alpha^2$. Thus, the fixed field $K_{H_3} = \{x \in K \mid \mu\rho(x) = x\}$ is given by $\mathbb{Q}(\alpha^2)$.
- K_{H_4} : Computing $\mu\rho^2(\gamma)$, the only unchanged term is $a + c\alpha^2 - ei - g\alpha^2 i$. Thus, the fixed field $K_{H_4} = \{x \in K \mid \mu\rho^2(x) = x\}$ is given by $\mathbb{Q}(\alpha^2, i)$.

□

Exercise 47.9(i). Express the symmetric function $y_1^2 + y_2^2 + y_3^2$ in y_1, y_2, y_3 over $\mathbb{Q}$ as a rational function of the elementary symmetric functions s_1, s_2 , and s_3 .

Solution. We start by computing s_1^2 :

$$s_1^2 = (y_1 + y_2 + y_3)^2 = y_1^2 + y_2^2 + y_3^2 + 2(y_1y_2 + y_1y_3 + y_2y_3) = y_1^2 + y_2^2 + y_3^2 + 2s_2.$$

Thus, we have $y_1^2 + y_2^2 + y_3^2 = s_1^2 - 2s_2$.

□

Exercise 47.10. Let α_1, α_2 , and α_3 be the zeros in $\mathbb{C}$ of the polynomial

$$x^3 - 4x^2 + 6x - 2 \in \mathbb{Q}[x].$$

Find the polynomial having as zeros precisely the following:

(i) $\alpha_1 + \alpha_2 + \alpha_3$.

(ii) α_1^2, α_2^2 , and α_3^2 .

Solution to (i). The polynomial having $\alpha_1 + \alpha_2 + \alpha_3$ as a zero is given by $x - (\alpha_1 + \alpha_2 + \alpha_3)$. We start by computing $\alpha_1 + \alpha_2 + \alpha_3$. By Viète's formulas, we have $\alpha_1 + \alpha_2 + \alpha_3 = 4$. Thus, the polynomial having $\alpha_1 + \alpha_2 + \alpha_3$ as a zero is given by $x - 4$.

□

Solution to (ii). The polynomial having α_1^2, α_2^2 , and α_3^2 as zeros is given by $(x - \alpha_1^2)(x - \alpha_2^2)(x - \alpha_3^2)$. We start by computing the elementary symmetric functions of α_1^2, α_2^2 , and α_3^2 :

$$\begin{aligned} s_1 &= \alpha_1^2 + \alpha_2^2 + \alpha_3^2 = (\alpha_1 + \alpha_2 + \alpha_3)^2 - 2(\alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3) = 4^2 - 2(6) = 16 - 12 = 4, \\ s_2 &= \alpha_1^2\alpha_2^2 + \alpha_1^2\alpha_3^2 + \alpha_2^2\alpha_3^2 = (\alpha_1\alpha_2)^2 + (\alpha_1\alpha_3)^2 + (\alpha_2\alpha_3)^2 = 6^2 - 4(4)(-2) = 36 + 32 = 68, \\ s_3 &= \alpha_1^2\alpha_2^2\alpha_3^2 = (\alpha_1\alpha_2\alpha_3)^2 = (-2)^2 = 4. \end{aligned}$$

Thus, the polynomial having α_1^2, α_2^2 , and α_3^2 as zeros is given by $x^3 - s_1x^2 + s_2x - s_3 = x^3 - 4x^2 + 68x - 4$. □

Exercise 47.11. Show that every finite group is isomorphic to some Galois group $G(K/F)$ for some finite normal extension K of some field F .

Solution. Let G be a finite group. By Cayley's theorem, G is isomorphic to a subgroup of S_n for some n , so it suffices to show every subgroup of S_n arises as a Galois group.

Let $F = \mathbb{Q}$ and let K be the splitting field of $f(x) = \prod_{i=1}^n (x - t_i)$ over the field $\mathbb{Q}(t_1, \dots, t_n)$ of rational functions in n indeterminates. By a standard result, the Galois group of this generic polynomial is S_n . However, it is cleaner to argue directly as follows.

Let $H \leq S_n$ be any subgroup (with $G \cong H$). Let $L = \mathbb{Q}(t_1, \dots, t_n)$ be a purely transcendental extension of $\mathbb{Q}$, on which S_n acts by permuting the indeterminates t_i . Let $F = L^{S_n}$ be the fixed field of S_n , which is the field of symmetric rational functions, and let $K = L^H$ be the fixed field of H . Since L/F is a finite Galois extension with $G(L/F) \cong S_n$, the Galois correspondence gives

$$G(L/K) \cong H,$$

and L/K is a finite normal extension. Setting $K' = K$ and $F' = L$ (relabeling), we have exhibited $H \cong G(L/K)$ as a Galois group of a finite normal extension. Since $G \cong H$, the group G is isomorphic to the Galois group $G(L/K)$. $\square$

Exercise 48.2. Classify the group of the polynomial $(x^{20} - 1) \in \mathbb{Q}[x]$ over $\mathbb{Q}$ according to the Fundamental Theorem of Finitely Generated Abelian Groups. Theorem 9.12. [Hint: Use Theorem 48.4.]

Solution. The splitting field of $x^{20} - 1$ over $\mathbb{Q}$ is the 20th cyclotomic extension $K = \mathbb{Q}(\zeta)$, where $\zeta = e^{2\pi i/20}$ is a primitive 20th root of unity. By Theorem 48.4, the Galois group $G(K/\mathbb{Q})$ is isomorphic to the group of positive integers less than 20 and relatively prime to 20 under multiplication modulo 20, which is $\mathbb{Z}_{20}^*$, and has order $\varphi(20)$.

We compute $\varphi(20) = \varphi(4)\varphi(5) = 2 \cdot 4 = 8$. The units of $\mathbb{Z}_{20}$ are

$$\mathbb{Z}_{20}^* = \{1, 3, 7, 9, 11, 13, 17, 19\}.$$

To classify this group, we use the factorization $20 = 4 \cdot 5$ and the Chinese Remainder Theorem, which gives

$$\mathbb{Z}_{20}^* \cong \mathbb{Z}_4^* \times \mathbb{Z}_5^*.$$

Now $\mathbb{Z}_4^* = \{1, 3\} \cong \mathbb{Z}_2$ and $\mathbb{Z}_5^* \cong \mathbb{Z}_4$ (since $\mathbb{Z}_5^*$ is cyclic of order $\varphi(5) = 4$). Therefore

$$G(K/\mathbb{Q}) \cong \mathbb{Z}_{20}^* \cong \mathbb{Z}_2 \times \mathbb{Z}_4.$$

This is the classification according to the Fundamental Theorem of Finitely Generated Abelian Groups: $G(K/\mathbb{Q})$ is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_4$. $\square$

Exercise 48.5. Find the smallest angle of integral degree, that is, $1^\circ, 2^\circ, 3^\circ$, and so on, constructible with a straightedge and a compass. [Hint: Constructing a 1° angle amounts to constructing the regular 360-gon, and so on.]

Solution. Constructing an angle of k° is equivalent to constructing the regular $(360/k)$ -gon, which is possible only if $\varphi(360/k)$ is a power of 2. We seek the largest k dividing 360 for which this holds, as that gives the smallest constructible integral angle.

The divisors of 360 corresponding to small angles are $n = 360/k$. By the theory of Section 48, a regular n -gon is constructible only if $\varphi(n)$ is a power of 2, which requires that every odd prime dividing n is a Fermat prime appearing to the first power. The Fermat primes are 3, 5, 17, 257, 65537.

We factor $360 = 2^3 \cdot 3^2 \cdot 5$. For $n \mid 360$, we need $\varphi(n)$ to be a power of 2, so n may contain no odd prime to a power greater than 1. Since $3^2 \mid 360$, any n divisible by 9 fails. Thus n must be of the form $2^a \cdot 3^b \cdot 5^c$

with $b, c \in \{0, 1\}$, i.e., $n \in \{1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 24, 60\}$ restricted to divisors of 360 with no squared odd prime factor. The largest such divisor of 360 is $n = 2^3 \cdot 3 \cdot 5 = 120$, giving

$$\varphi(120) = \varphi(8)\varphi(3)\varphi(5) = 4 \cdot 2 \cdot 4 = 32 = 2^5,$$

which is indeed a power of 2, so the regular 120-gon is constructible. This corresponds to an angle of

$$k = \frac{360}{120} = 3^\circ.$$

For $k = 2^\circ$ we would need $n = 180 = 2^2 \cdot 3^2 \cdot 5$, but $3^2 \nmid 180$ so $\varphi(180) = \varphi(4)\varphi(9)\varphi(5) = 2 \cdot 6 \cdot 4 = 48$, which is not a power of 2, so the regular 180-gon is not constructible. For $k = 1^\circ$ we would need $n = 360 = 2^3 \cdot 3^2 \cdot 5$, and similarly $3^2 \nmid 360$ forces $\varphi(360) = 96$, not a power of 2.

Therefore the smallest constructible integral angle is 3° . $\square$

Exercise 48.8. Show that

$$x^n - 1 = \prod_{d|n} \Phi_d(x),$$

in $\mathbb{Q}[x]$, where the product is over all divisors d of n .

Solution. Recall that $\Phi_d(x)$ is the d th cyclotomic polynomial, whose roots are exactly the primitive d th roots of unity. We show the two sides have the same roots with the same multiplicities, and the same leading coefficient.

The roots of $x^n - 1$ in $\mathbb{C}$ are exactly the n th roots of unity $\{\zeta \in \mathbb{C} \mid \zeta^n = 1\}$, each occurring with multiplicity 1. Every n th root of unity ζ is a primitive d th root of unity for exactly one divisor $d \mid n$, namely $d = \text{ord}(\zeta)$, which divides n . Conversely, every primitive d th root of unity for any $d \mid n$ is an n th root of unity. Thus the roots of $\prod_{d|n} \Phi_d(x)$ are exactly the primitive d th roots of unity ranging over all $d \mid n$, which is precisely the set of all n th roots of unity, each appearing exactly once. Hence $x^n - 1$ and $\prod_{d|n} \Phi_d(x)$ have the same roots in $\mathbb{C}$, each with multiplicity 1.

Since both sides are monic polynomials of degree n (as $\sum_{d|n} \varphi(d) = n$) with the same roots, we conclude

$$x^n - 1 = \prod_{d|n} \Phi_d(x)$$

in $\mathbb{C}[x]$, and in particular in $\mathbb{Q}[x]$. $\square$

Exercise 48.10. Find $\Phi_{12}(x)$ in $\mathbb{Q}[x]$. [Hint: Use Exercises 8 and 9.]

Solution. By Exercise 8 (Problem 48.8), $x^{12} - 1 = \prod_{d|12} \Phi_d(x)$. The divisors of 12 are 1, 2, 3, 4, 6, 12, so

$$x^{12} - 1 = \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_4(x)\Phi_6(x)\Phi_{12}(x).$$

From Exercise 9, the cyclotomic polynomials for small n are:

$$\Phi_1(x) = x - 1, \quad \Phi_2(x) = x + 1, \quad \Phi_3(x) = x^2 + x + 1, \quad \Phi_4(x) = x^2 + 1, \quad \Phi_6(x) = x^2 - x + 1.$$

The product of the known factors is

$$\Phi_1\Phi_2\Phi_3\Phi_4\Phi_6 = (x - 1)(x + 1)(x^2 + x + 1)(x^2 + 1)(x^2 - x + 1).$$

We compute step by step. First, $(x - 1)(x + 1) = x^2 - 1$. Next,

$$(x^2 - 1)(x^2 + x + 1) = x^4 + x^3 + x^2 - x^2 - x - 1 = x^4 + x^3 - x - 1.$$

Then,

$$(x^4 + x^3 - x - 1)(x^2 + 1) = x^6 + x^5 - x^3 - x^2 + x^4 + x^3 - x - 1 = x^6 + x^5 + x^4 - x^2 - x - 1.$$

Finally,

$$(x^6 + x^5 + x^4 - x^2 - x - 1)(x^2 - x + 1).$$

Expanding:

$$\begin{aligned} x^2(x^6 + x^5 + x^4 - x^2 - x - 1) &= x^8 + x^7 + x^6 - x^4 - x^3 - x^2, \\ -x(x^6 + x^5 + x^4 - x^2 - x - 1) &= -x^7 - x^6 - x^5 + x^3 + x^2 + x, \\ 1 \cdot (x^6 + x^5 + x^4 - x^2 - x - 1) &= x^6 + x^5 + x^4 - x^2 - x - 1. \end{aligned}$$

Summing by degree:

$$x^8 + (1-1)x^7 + (1-1+1)x^6 + (0-1+1)x^5 + (-1+1+1)x^4 + (-1+1)x^3 + (-1+1-1)x^2 + (1-1)x - 1,$$

which simplifies to $x^8 + x^6 + x^4 - x^2 - 1$. But this has degree 8, whereas $\deg \Phi_{12} = \varphi(12) = 4$ and $\deg(x^{12} - 1) = 12$, so we should have $8 + 4 = 12$. Solving for $\Phi_{12}(x)$:

$$\Phi_{12}(x) = \frac{x^{12} - 1}{x^8 + x^6 + x^4 - x^2 - 1}.$$

Performing polynomial long division of $x^{12} - 1$ by $x^8 + x^6 + x^4 - x^2 - 1$ yields

$$\Phi_{12}(x) = x^4 - x^2 + 1. \quad \square$$

Exercise 48.12. Let $n, m \in \mathbb{Z}^+$ be relatively prime. Show that the splitting field in $\mathbb{C}$ of $x^{nm} - 1$ over $\mathbb{Q}$ is the same as the splitting field in $\mathbb{C}$ of $(x^n - 1)(x^m - 1)$ over $\mathbb{Q}$.

Solution. Let $\zeta = e^{2\pi i/nm}$, a primitive nm th root of unity, so that the splitting field of $x^{nm} - 1$ over $\mathbb{Q}$ is $K = \mathbb{Q}(\zeta)$. Let L be the splitting field of $(x^n - 1)(x^m - 1)$ over $\mathbb{Q}$. The roots of $(x^n - 1)(x^m - 1)$ are the n th and m th roots of unity, so $L = \mathbb{Q}(\zeta_n, \zeta_m)$ where $\zeta_n = e^{2\pi i/n}$ and $\zeta_m = e^{2\pi i/m}$ are primitive n th and m th roots of unity respectively.

Since $\gcd(n, m) = 1$, by Bezout's theorem there exist $s, t \in \mathbb{Z}$ such that $sn + tm = 1$. Then

$$\zeta = e^{2\pi i/nm} = e^{2\pi i(sn+tm)/nm} = e^{2\pi is/m} \cdot e^{2\pi it/n} = \zeta_m^s \cdot \zeta_n^t \in L.$$

Hence $K = \mathbb{Q}(\zeta) \subseteq L$.

The n th roots of unity are nm th roots of unity (since $(\zeta^m)^n = \zeta^{mn} = 1$), so all roots of $x^n - 1$ lie in K . Similarly all roots of $x^m - 1$ lie in K . Hence $L \subseteq K$.

Therefore $K = L$, and the two splitting fields coincide. $\square$